
Please review and study the solutions for homework assignment 1 and 2, and exercise problems before proceeding with this assignment.

Homework 3 (CTL): ComS/CprE/SE 412, ComS 512

Due-date: Mar 25 at 11:59PM.

Submission format. Your submission file must be pdf format. You will use canvas-gradescope for submission (this will allow you to associate scanned pages/pdfs of your submission with the question number). Here is a short video on how to submit homework using gradescope

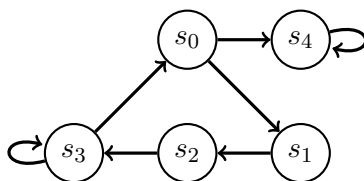
<https://www.youtube.com/watch?v=u-pK4GzpId0>

You are encouraged to use latex to prepare your submission but this is not a requirement.

1. The ATM controller program must satisfy the following properties. Express these properties in CTL.
 - (a) Whenever there is a successful authentication, the ATM always eventually either dispenses cash or receives a cancel transaction request.
 - (b) the ATM never dispenses cash after cancel transaction request if there is not successful authentication in between.
2. A self-driving car on complete autonomy satisfies the following properties. Express these properties in CTL.
 - (a) From any configuration, the autonomous system can eventually enter a safestate.
 - (b) The autonomous system disables itself if it brings the car to a configuration which always leads to unsafe states.

(5+5)

3. For the following Kripke structure¹ with $b \in L(s_3)$, $\{a, b\} \subseteq L(s_2)$, $a \in L(s_1)$ and $b \in L(s_4)$,



We are given a function defined as follows:

$$A_\varphi(Z) = [\varphi]_M \cap R_\forall(R_\forall(Z))$$

where φ is some CTL formula, $[\varphi]_M$ is the semantics of φ (in the context of a Kripke structure $M = (S, T, L)$) defined as a set of states of a Kripke structure, and $R_\forall$ is

$$R_\forall(Z) = \{s \mid \forall s'. (s, s') \in T \Rightarrow s' \in Z\}$$

¹For the variables $x \in \{a, b\}$, if we do not write $x \in L(s)$, then $x \notin L(s)$.

- (a) Compute the greatest fixed point A_b for the given Kripke structure.
- (b) **512 only** It is claimed that the semantics of the property expressed by the greatest fixed point of A_b cannot be expressed in CTL. Justify the validity of the claim.

(6+4(Extra Credit))

4. Let f, g, h be monotonically non-decreasing set functions

$$f(Z) = g(Z) \cup h(Z)$$

- (a) Provide example definitions for f, g, h such that LFP of f is the union of LFP of g and h . Justify your answer.
- (b) Provide example definitions for f, g, h such that LFP of f is not the union of LFP of g and h . Justify your answer.

Consider using some CTL (non)equivalences and their fixed point characterizations for answering the problem.

(5+5)

For the following ROBDD problems, you do not need to show the edges from the nodes that go to the terminal node 0.

5. Draw the ROBDD for the following function.

$$f(x_1, x_2, x_3, x_4) = \begin{cases} 1 & \text{if the sum of } x_i\text{'s is equal to even} \\ 0 & \text{otherwise} \end{cases}$$

In the above, the domain of x_i is $\{0, 1\}$. (5)

6. Consider the boolean function which captures the binary addition of two binary numbers as follows:

$$\begin{array}{r} x_1 \quad x_0 \\ + \quad y_1 \quad y_0 \\ \hline z_2 \quad z_1 \quad z_0 \end{array}$$

The boolean function evaluates to true if and only if the correct semantics of addition is realized. For instance,

$$\begin{array}{r} x_1 = 0 \quad x_0 = 1 \\ + \quad y_1 = 1 \quad y_0 = 1 \\ \hline z_2 = 1 \quad z_1 = 0 \quad z_0 = 0 \end{array}$$

Or,

$$\begin{array}{r} x_1 = 1 \quad x_0 = 1 \\ + \quad y_1 = 1 \quad y_0 = 1 \\ \hline z_2 = 1 \quad z_1 = 1 \quad z_0 = 0 \end{array}$$

For the smallest size ROBDD representing the above function, for each variable, write the number of nodes. That is, how many nodes will be labeled with x_0 , how many nodes will be labeled with x_1 and so on. Justify your solution. (5)